

# Simulating Spectral Sampling of the MICA Files with the Enfys IR Spectrometer for the ExoMars Rosalind Franklin Rover

February 21, 2025

## 1 Abstract

Here we investigate the expected ability of a new remote-sensing stand-off infrared spectrometer, *Enfys*, for the ExoMars Rosalind Franklin Rover, to resolve diagnostic features of the reflectance spectra of key minerals previously identified on the surface of Mars.

## 2 Introduction

### 2.1 Background

The ESA ExoMars Rosalind Franklin Rover is planned to land at the Oxia Planum region of Mars <sup>1</sup> to characterise the geochemical and aqueous history of the environment and near subsurface (<2 m), and search for signs of past or present life <sup>2</sup>.

Previously the rover payload included a ROSCOSMOS (Russian) led instrument, the Infrared Spectrometer for ExoMars (ISEM) <sup>3</sup>. Following the geopolitical fallout of the Russian invasion of Ukraine in 2021 the ISEM instrument was removed from the rover. A new UK-led instrument, *Enfys* (Welsh for ‘rainbow’), has been approved at the ESA ministerial level to replace it.

The purpose of the spectrometer is to provide high-spectral resolution observations of points of interest found in panoramic colour and multispectral image products, captured by the mast-mounted PanCam instrument <sup>4</sup> of the rover. The short-wave infrared range is sensitive to the IR-active H<sub>2</sub>O, OH, and CO<sub>3</sub> molecules found in the carbonates and hydrated clays, sulphates and silicates that

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<sup>1</sup>Quantin-Nataf, C., Carter, J., Mandon, L., Thollot, P., Balme, M., Volat, M., Pan, L., Loizeau, D., Millot, C., Breton, S., Dehouck, E., Fawdon, P., Gupta, S., Davis, J., Grindrod, P. M., Pacifici, A., Bultel, B., Allemand, P., Ody, A., ... Broyer, J. (2021). Oxia Planum: The Landing Site for the ExoMars “Rosalind Franklin” Rover Mission: Geological Context and Prelanding Interpretation. *Astrobiology*, 21(3). <https://doi.org/10.1089/ast.2019.2191>

<sup>2</sup>Vago, J. L., Westall, F., Pasteur Instrument Teams, Landing S, Coates, A. J., Jaumann, R., Korablev, O., Ciarletti, V., Mitrofanov, I., Josset, J.-L., De Sanctis, M. C., Bibring, J.-P., Rull, F., Goesmann, F., Steininger, H., Goetz, W., Brinckerhoff, W., Szopa, C., Raulin, F., Westall, F., ... the ExoMars Project Team. (2017). Habitability on Early Mars and the Search for Biosignatures with the ExoMars Rover. *Astrobiology*, 17(6–7), 471–510. <https://doi.org/10.1089/ast.2016.1533>

<sup>3</sup>Korablev, O. I., Dobrolensky, Y., Evdokimova, N., Fedorova, A. A., Kuzmin, R. O., Mantsevich, S. N., Cloutis, E. A., Carter, J., Poulet, F., Flahaut, J., Griffiths, A., Gunn, M., Schmitz, N., Martín-Torres, J., Zorzano, M.-P., Rodionov, D. S., Vago, J. L., Stepanov, A. V., Titov, A. Yu., ... Ivanov, A. Yu. (2017). Infrared Spectrometer for ExoMars: A Mast-Mounted Instrument for the Rover. *Astrobiology*, 17(6–7), 542–564. <https://doi.org/10.1089/ast.2016.1543>

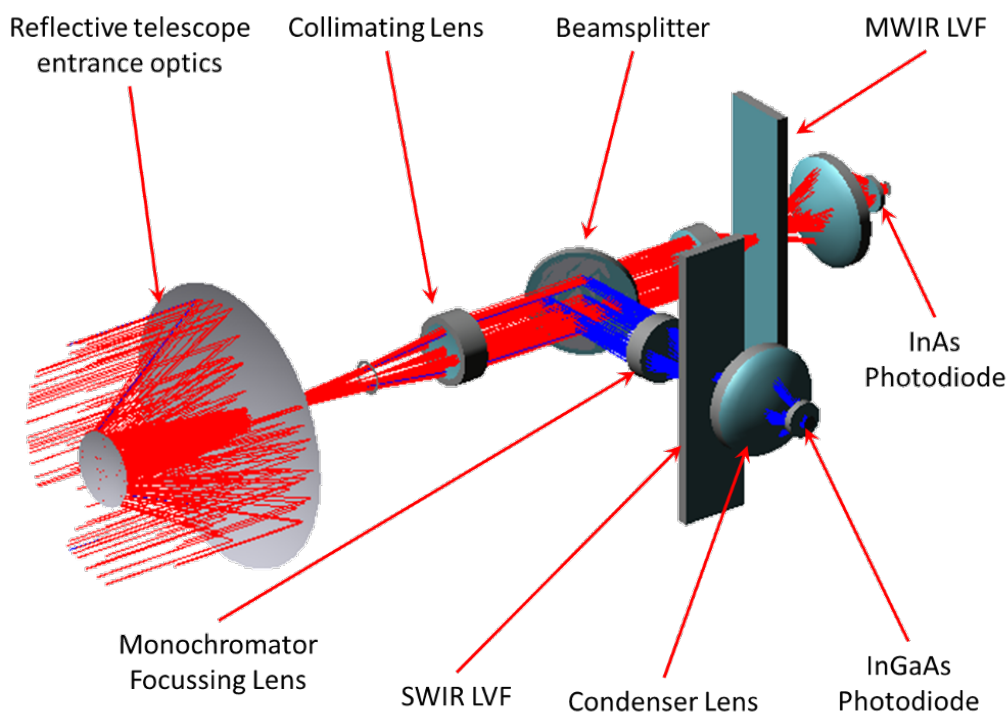
<sup>4</sup>Coates, A. J., Jaumann, R., Griffiths, A. D., Leff, C. E., Schmitz, N., Josset, J.-L., Paar, G., Gunn, M., Hauber, E., Cousins, C. R., Cross, R. E., Grindrod, P., Bridges, J. c., Balme, M., Gupta, S., Crawford, I. A., Irwin, P., Stabbins, R., Tirsch, D., ... the PanCam Team. (2017). The PanCam Instrument for the ExoMars Rover. *Astrobiology*, 17(6–7), 511–541. <https://doi.org/10.1089/ast.2016.1548>

have been found previously on Mars, and that give insights to the geochemical and aqueous history of the environment. The high spectral resolution is required to resolve the subtle changes in the IR-active molecule band depths, positions and asymmetries that are induced by the presence of the various other elements and molecules in different mineral groups, species and phases.

## 2.2 Enfys

The *Enfys* infrared spectrometer uses a scanning linear-variable-filter (LVF) optical design to observe incident narrowband intensity in the spectral range of 0.9 - 3.1  $\mu\text{m}$  with a spectral resolving power of  $\sim 100$ , and with an expected signal-to-noise ratio of  $>2000$  at 0.9  $\mu\text{m}$  and  $>100$  at 3.1  $\mu\text{m}$ . The spectral range is covered by two photodiodes and two separate linear-variable filters; a 0.9 - 1.7  $\mu\text{m}$  range filter is paired with an uncooled InGaAs photodiode, and the 1.6 - 3.1  $\mu\text{m}$  range filter is paired with a cooled TE photodiode.

```
[1]: from IPython.display import display, Image, HTML
      display(Image('enfys_diagram.png'))
```



**Figure 1** Schematic of the *Enfys* IR spectrometer optical path.

## 2.3 Problem

This optical design is functionally different to the Acousto-Optic Tunable Filter design of ISEM, so it is important that the design is demonstrated to fulfil the performance requirements of the instrument, and expectations of performance can be set prior to laboratory tests of the completed instrument.

At time of writing, the spectrometer is in the development stage. This notebook describes how the Spectral Parameters Toolkit [ref] can be used to convert the developing expectations of the spectral range and resolution and radiometric sensitivity of the instrument into predictions of its ability to discriminate mineral species and phases on the Mars surface at the rover landing site.

### 3 Method

To investigate the expected performance we will use high-resolution laboratory measured spectra of a range of minerals expected at the Mars surface, and computationally simulate the sampling of the spectra with the expected spectral response of *Enfys*. We will perform repeat sampling with synthetically added noise to investigate the spectral and radiometric limits of the instrument.

We will produce a number of visualisations to investigate the predicted instrument performance, and demonstrate the limits of the instrument discriminatory ability.

#### 3.1 Spectral Resampling

We take a simple approach to resampling, as follows.

We neglect the introduction of reflectance calibration uncertainty, and assume that the spectral transmission of the linear-variable filter movement is perfectly calibrated such that a measurement at central wavelength ( $\lambda_{cwl}$ ) can be requested with arbitrary precision.

We assume an observation of  $\lambda_{cwl}$  has a Cauchy transmission profile, and, prior to component characterisation, a spectral resolving power of  $\lambda/\Delta\lambda = 100$  is achieved for all  $\lambda_{cwl}$  [ref], such that the Cauchy transmission profile for  $\lambda_{cwl}$  has a full-width-at-half-maximum of  $\Delta\lambda = \lambda_{cwl}/100$ .

The Cauchy transmission profile  $T(\lambda|\lambda_{cwl})$  is given by

$$T(\lambda|\lambda_{cwl}) = \frac{\gamma^2}{(\lambda - \lambda_{cwl})^2 + \gamma^2}$$

where  $\gamma = \Delta\lambda/2 = \lambda_{cwl}/100/2$ .

We assume that a high-resolution laboratory spectrum approximates the continuous function  $R(\lambda)$ , such that the calibrated reflectance at  $\lambda_{cwl}$  has the expected value  $R[\lambda_{cwl}]$  of:

$$R[\lambda_{cwl}] = \frac{\int R(\lambda)T(\lambda|\lambda_{cwl})d\lambda}{\int T(\lambda|\lambda_{cwl})d\lambda}$$

To simulate noise we add  $\varepsilon$ , an additive noise term that we draw from the Gaussian distribution with variance:

$$\sigma^2 = \frac{\sqrt{R[\lambda_{cwl}]}}{\text{SNR}}$$

under the assumption that maximum SNR is only achieved for  $R = 1$ , and the noise is dominated by counting noise, such that  $\sigma$  follows a Poissonian relationship with the signal  $R$ .

For each observation  $R[\lambda_{cwl}]$  we draw 100 repeat samples, such that

$$R_i[\lambda_{cwl}] = R[\lambda_{cwl}] + \varepsilon_i$$

with

$$\varepsilon_i \sim \mathcal{N}\left(0, \frac{\sqrt{R[\lambda_{cwl}]}}{\text{SNR}}\right)$$

## 3.2 Analysis

For now, we simply produce plots of the resampled spectra. We will include more details of analysis as that is developed here.

## 3.3 Notebook and Environment Setup

Here we include the required code blocks for setting up this notebook for this study.

```
[2]: %load_ext autoreload
      %autoreload 2
      %matplotlib inline
```

We will be using the Spectral Parameters Toolkit [ref]. First we import the required modules.

```
[3]: import sptk.config as cfg
      from sptk.material_collection import MaterialCollection
      from sptk.instrument import Instrument
      from sptk.observation import Observation
      from sptk.spectral_library_analyser import SpectralLibraryAnalyser as sla
```

We set our simulation resolution range to extend the range of Enfys, of 0.9 - 3.1  $\mu\text{m}$ , such that the long tails of the Cauchy distribution of the linear-variable-filter profiles can be accommodated during sampling. We leave our spectral resolution at 1 nm for all  $\lambda$ .

```
[4]: cfg.update_sample_res('wvl_min', 850)
      cfg.update_sample_res('wvl_max', 3300)
```

Finally we set the project name.

```
[5]: PROJECT_NAME = 'enfys_mica'
```

## 4 Data

For this study we are using a spectral library of minerals identified through CRISM analysis ([the MICA files](#))<sup>5</sup>. Here we use the laboratory spectra that have been identified that best represent the type localities presented in the MICA files. These can be accessed [here](#), and are sourced from the RELAB and USGS public spectral libraries.

We have grouped the spectral library into categories as follows, and included this categorisation in the sptk distribution, accessed from the `sptk.material_sets` module.

```
[6]: from sptk.material_sets import MICA_SET
```

```
[7]: MICA_SET
```

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<sup>5</sup>Viviano-Beck, C. E., Seelos, F. P., Murchie, S. L., Kahn, E. G., Seelos, K. D., Taylor, H. W., Taylor, K., Ehlmann, B. L., Wiseman, S. M., Mustard, J. F., & Morgan, M. F. (2014). Revised CRISM spectral parameters and summary products based on the currently detected mineral diversity on Mars. *Journal of Geophysical Research: Planets*, 119(6), 1403–1431. <https://doi.org/10.1002/2014JE004627>



```
[7]: {'iron oxides & primary silicates': [('hematite', '*'),
    ('forsterite', '*'),
    ('fayalite', '*'),
    ('plagioclase', '*'),
    ('orthopyroxene', '*'),
    ('clinopyroxene', '*')],
'sulfates': [('kieserite', '*'),
    ('alunite', '*'),
    ('jarosite', '*'),
    ('mg-sulphate', '*'),
    ('gypsum', '*'),
    ('bassanite', '*')],
'phyllosilicates': [('montmorillonite', '*'),
    ('kaolinite', '*'),
    ('margarite', '*'),
    ('illite', '*'),
    ('nontronite', '*'),
    ('saponite', '*'),
    ('talc', '*'),
    ('serpentine', '*'),
    ('chlorite', '*')],
'carbonates': [('magnesite', '*'), ('calcite', '*')],
'hydrated silicates & halides': [('siliceous-sinter', '*'),
    ('halite', '*'),
    ('epidote', '*'),
    ('analcime', '*'),
    ('prehnite', '*')]]}
```

We use the `sptk.material_collection` module to parse the spectral library:

```
[8]: matcol = MaterialCollection(
    MICA_SET,
    'MICA_new_dirtree',
    PROJECT_NAME,
    load_existing=False,
    balance_classes=False,
    random_bias_seed=None,
    allow_out_of_bounds=True,
    plot_profiles=False,
    export_df=True)
```

Building new `enfys_mica` `MaterialCollection` DF

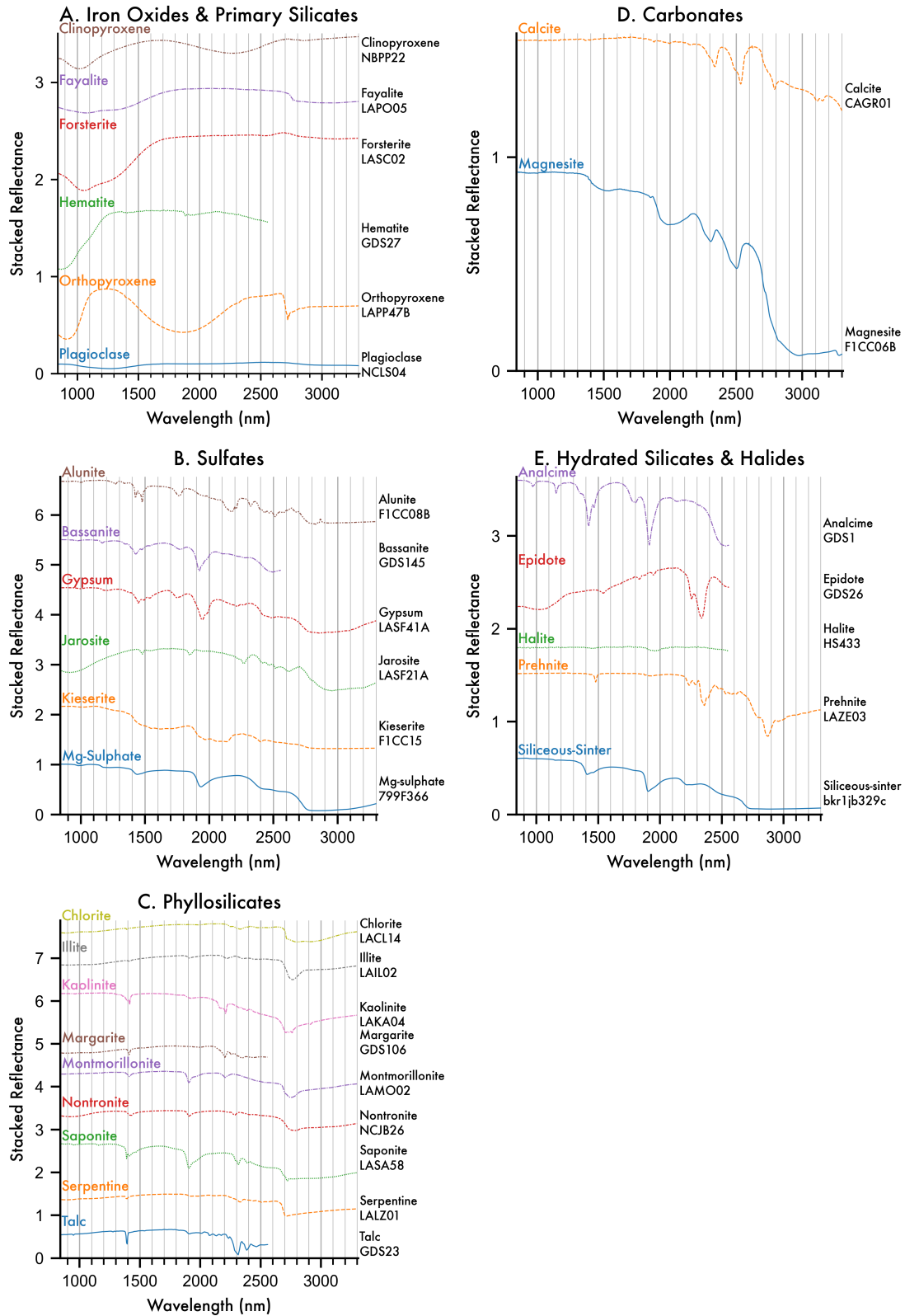
```
Iron Oxides & Primary Silicates: NBPP22.csv (Clinopyroxene) loaded
Iron Oxides & Primary Silicates: LAPP47B.csv (Orthopyroxene) loaded
Iron Oxides & Primary Silicates: LAP005.csv (Fayalite) loaded
Iron Oxides & Primary Silicates: LASC02.csv (Forsterite) loaded
Iron Oxides & Primary Silicates: GDS27.csv (Hematite) loaded
```

Iron Oxides & Primary Silicates: NCLS04.csv (Plagioclase) loaded  
Sulfates: F1CC08B.csv (Alunite) loaded  
Sulfates: GDS145.csv (Bassanite) loaded  
Sulfates: LASF41A.csv (Gypsum) loaded  
Sulfates: LASF21A.csv (Jarosite) loaded  
Sulfates: F1CC15.csv (Kieserite) loaded  
Sulfates: 799F366.csv (Mg-Sulphate) loaded  
Phyllosilicates: LACL14.csv (Chlorite) loaded  
Phyllosilicates: LAKA04.csv (Kaolinite) loaded  
Phyllosilicates: LAIL02.csv (Illite) loaded  
Phyllosilicates: GDS106.csv (Margarite) loaded  
Phyllosilicates: LALZ01.csv (Serpentine) loaded  
Phyllosilicates: LAM002.csv (Montmorillonite) loaded  
Phyllosilicates: NCJB26.csv (Nontronite) loaded  
Phyllosilicates: LASA58.csv (Saponite) loaded  
Phyllosilicates: GDS23.csv (Talc) loaded  
Carbonates: CAGR01.csv (Calcite) loaded  
Carbonates: F1CC06B.csv (Magnesite) loaded  
Hydrated Silicates & Halides: HS433.csv (Halite) loaded  
Hydrated Silicates & Halides: LAZE03.csv (Prehnite) loaded  
Hydrated Silicates & Halides: bkrijb329c.csv (Siliceous-Sinter) loaded  
Hydrated Silicates & Halides: GDS26.csv (Epidote) loaded  
Hydrated Silicates & Halides: GDS1.csv (Analcime) loaded  
Loading 28 of entries complete.  
Exporting the Material Collection to CSV and Pickle formats...

We can use the `material_collection` plotting routines to visualise the library.

```
[9]: ax = matcol.plot_profiles(stacked=True, scope='categories', groupby='Species')
```

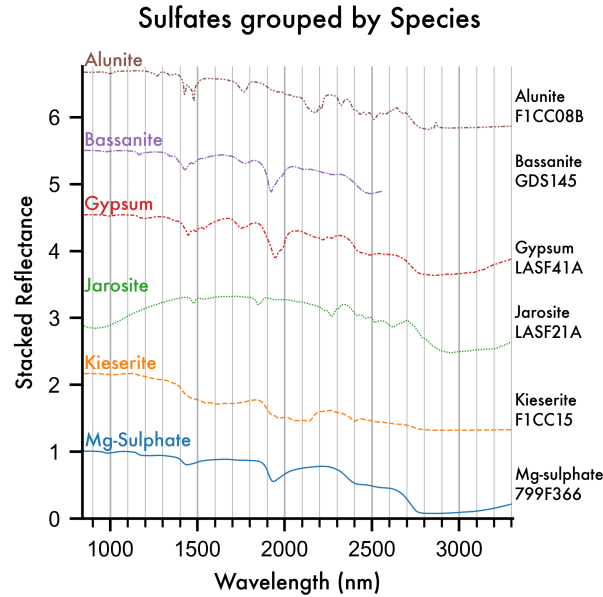
## Categories grouped by Species



**Figure 2** High resolution MICA Files spectral library, arranged by category and grouped by mineral species.

```
[10]: matcol.plot_profiles(scope='sulfates', groupby='Species', stacked=True)
```

```
[10]: [<AxesSubplot: xlabel='Wavelength (nm)', ylabel='Stacked Reflectance'>]
```



We can see that not all of the entries cover the complete 0.9 - 3.1  $\mu\text{m}$  range of *Enfys*, so in future work we should look to replace these MICA files laboratory type spectra with spectra that all span the entire range.

## 5 Results

### 5.1 Simulating the Enfys Spectral Response

To simulate *Enfys*, rather than providing a file with the expected spectral response of each position of the Linear-Variable Filter (LVF, above), we specify the spectral range and resolution and sampling regime, and generate a set of central-wavelengths and bandwidths. We do this with the `InstrumentBuilder` class of the `sptk.instrument` module.

```
[11]: from sptk.instrument import InstrumentBuilder
```

The expected spectral resolution of *Enfys* is  $\lambda/\Delta\lambda = 100$ , and the range is 0.9 - 3.1  $\mu\text{m}$ .

```
[12]: enfys_builder = InstrumentBuilder(
    instrument_name='enfys',
    instrument_type='lvf',
```

```
sampling='nyquist',
resolution = 100,
spectral_range=[900, 3101],
snr_range = [2000, 100])
```

Exporting instrument to ../data/instruments/enfys.csv...

```
[13]: from sptk.instrument import Instrument
```

Now we build the transmission profiles for each LVF position, according to the list of CWLs and FWHMs generated above. These are accessed by reading the .csv file produced by the InstrumentBuilder procedure.

When generating the profiles, we turn the CWL and FWHM into a profile by assuming a distribution function. For Enfys we use the Cauchy distribution, which behaves similarly to a Gaussian, but with wider ‘wings’. It is parameterised by the full-width-at-half-maximum, as the distribution strictly has no finite variance. We have implemented a `build_cauchy_filter` method for the `Instrument` class.

```
[14]: enfys = Instrument(
    name = 'enfys', # filename of the instrument information, held in /
    ↪software/data/instruments/
    project_name = matcol.project_name, # the project directory name,
    ↪hosting outputs in the /spectral_parameter_studies/ directory
    shape = 'cauchy',
    load_existing=False, # if True, load existing instrument data from
    ↪the project directory
    plot_profiles=False, # if true, plot the transmission profiles
    ↪during construction
    export_df=True) # if True, export the instrument transmission
    ↪profiles to the project directory
```

Building Instrument...

Building new DataFrame for 'enfys'...

Exporting the Instrument to CSV and Pickle formats...

```
[15]: fig ,ax = enfys.plot_instrument_characteristics()
```

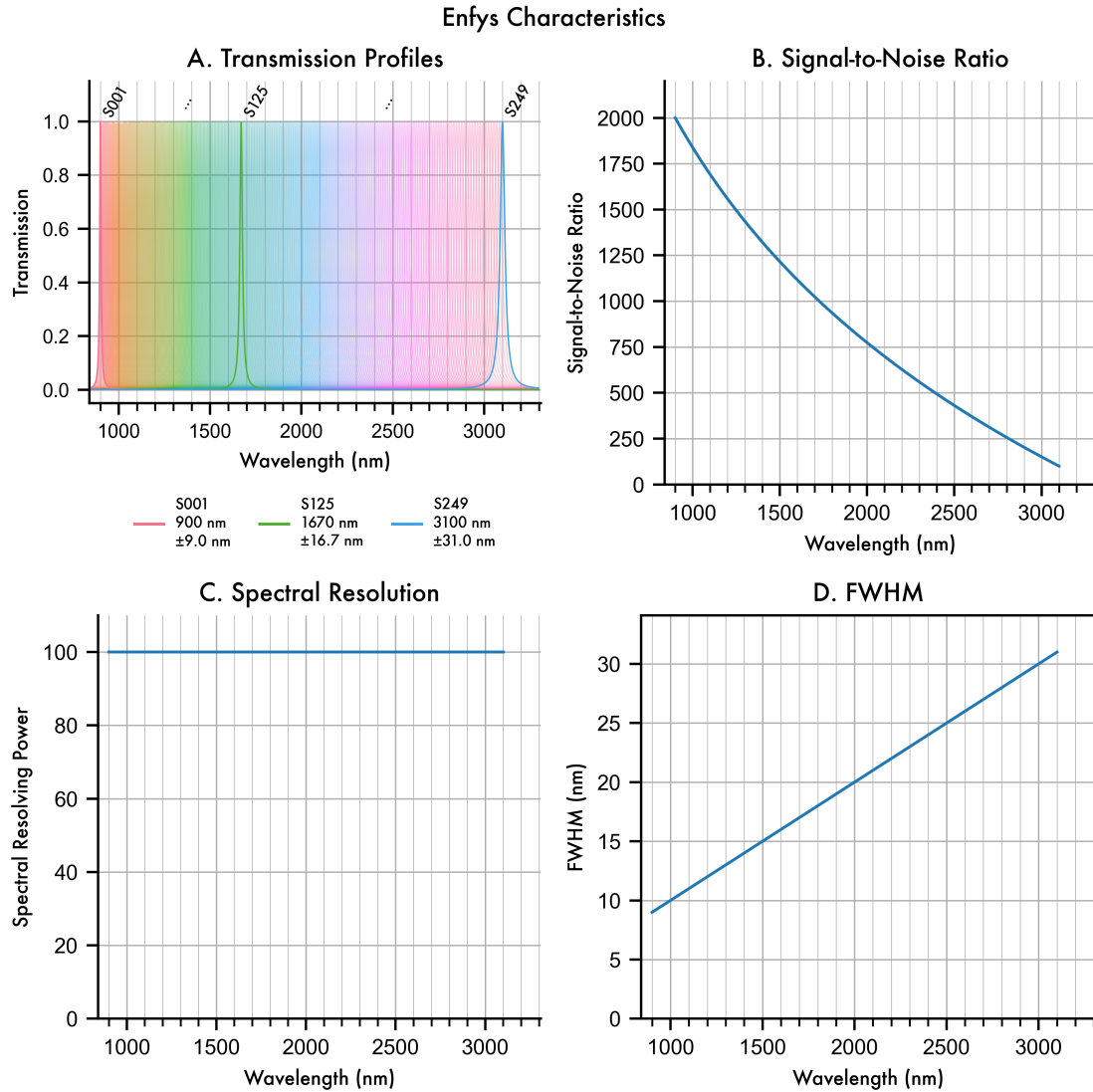
Plotting Instrument Transmission...

Plotting Signal-to-Noise Ratio...

Plotting Spectral Resolution...

Plotting FWHM...

Plots exported to ../projects/enfys\_mica/instrument



**Figure 3** Simulated spectral response and resolution of *Enfys*, based on preliminary expectations of the spectral and radiometric response.

## 5.2 Sampling MICA with Enfys

We now make an `Observation` object by sampling the MICA files `MaterialCollection` with the `Enfys Instrument`.

```
[16]: from sptk.observation import Observation
```

```
[17]: obs = Observation(
    material_collection=matcol,
    instrument = enfys,
    load_existing=False,
```

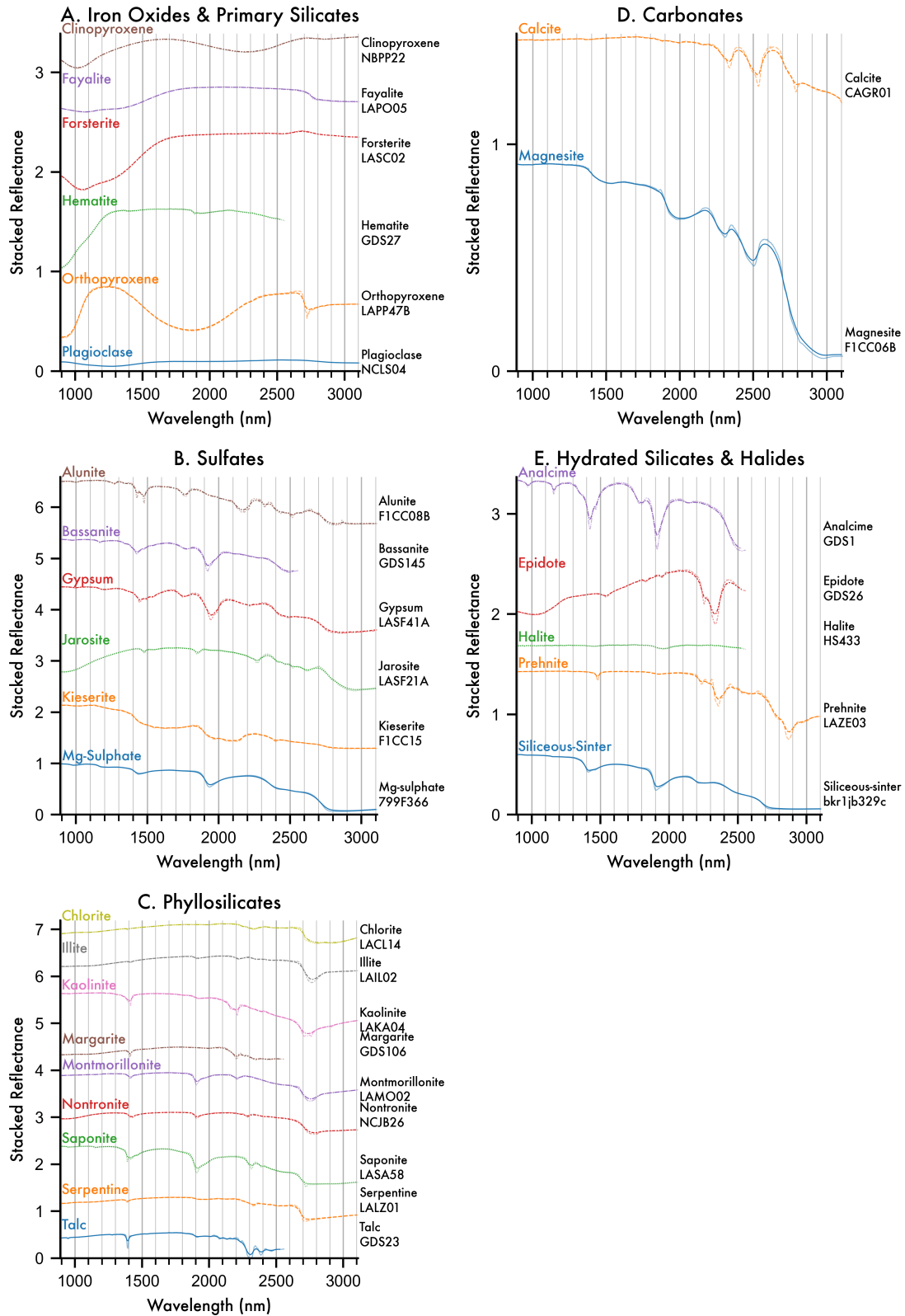
```
plot_profiles=False,  
export_df=True)
```

```
Building new Observation DataFrame  
for 'enfys_mica'...  
Sampling the Material Collection with the Instrument...  
Sampling complete.  
Exporting the Observation Pickle format...  
Observation export complete.
```

We can plot the sampled spectra against the input high-spectra below:

```
[18]: axes = obs.plot_profiles(scope='categories', groupby='Species', stacked=True,  
    ↪ hires_under=True)
```

## Enfys Sampled Categories grouped by Species



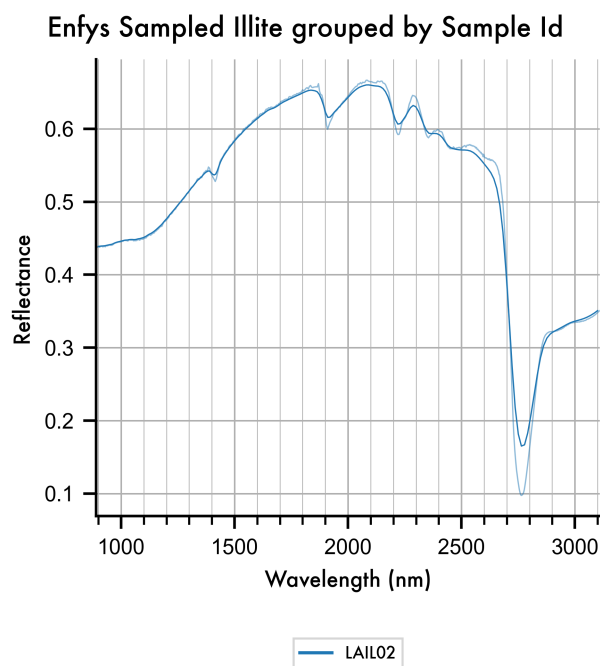


**Figure 4** The MICA files as sampled by *Enfys*, with the original high-resolution spectra plotted underneath each.

We can now see where there are deviations of the *Enfys* sampled spectra from the high-resolution source spectra.

For example, illite, plotted below for clarity.

```
[19]: axes = obs.plot_profiles(scope='illite', groupby='Sample ID', stacked=False,
    ↪ hires_under=True)
```



**Figure 5** Deviation of the *Enfys* sampled spectrum from the high-resolution input spectrum for Illite.

### 5.3 Adding Synthetic Noise

As discussed in the method, we apply a simple additive Gaussian noise component, by redrawing  $n = 100$  additional measurements. We can plot the full range of the resultant samples

```
[20]: noise_obs = obs.add_noise(100)
```

/Users/sci/Documents/software/sptk/src/sptk/observation.py:220:

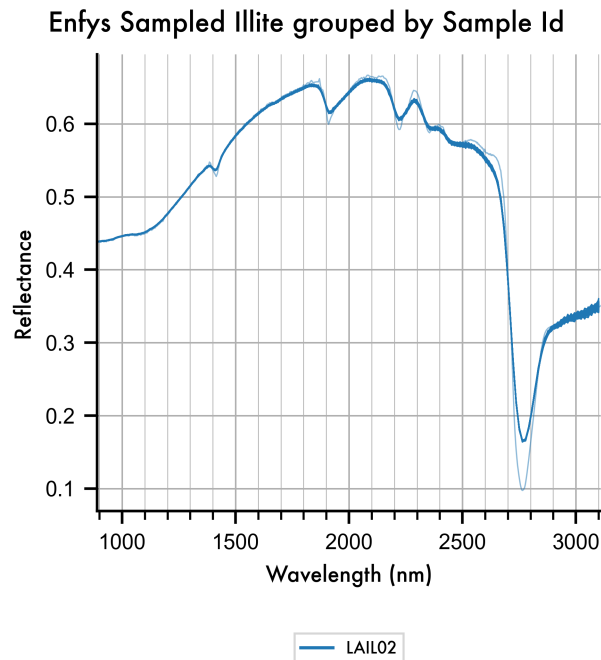
SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: <https://pandas.pydata.org/pandas->

```
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
obs_df[self.wvlsl].clip(lower = 0.0, inplace=True) # clip to range
```

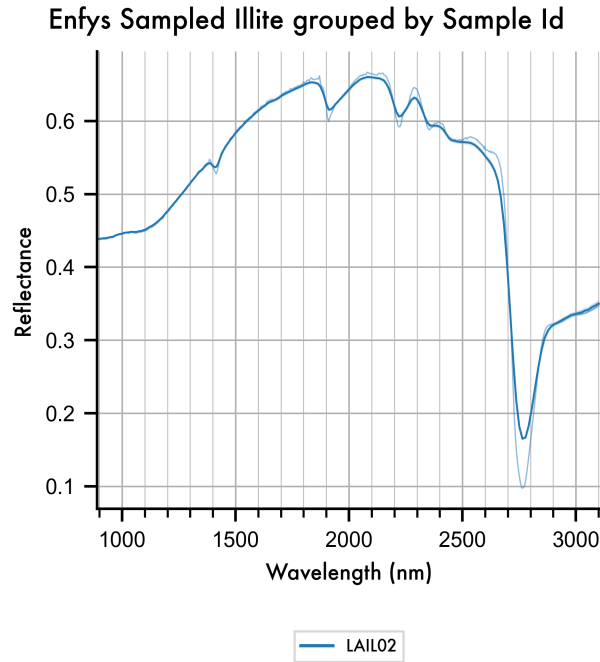
```
[21]: axes = obs.plot_profiles(scope='illite', groupby='Sample ID', stacked=False,
↳ hires_under=True, ci=False)
```



**Figure 6** Noisy resampled example spectrum for Illite, showing 100 repeat samples according to the spectral signal-to-noise ratio illustrated in figure 3.B.

We can also plot the mean observed line, with shading indicating the standard deviation of the noisy data, shown below in figure 7, where the ‘ci’ keyword has been applied (‘ci’: confidence-interval).

```
[22]: axes = obs.plot_profiles(scope='illite', groupby='Sample ID', stacked=False,
↳ hires_under=True, ci=True)
```



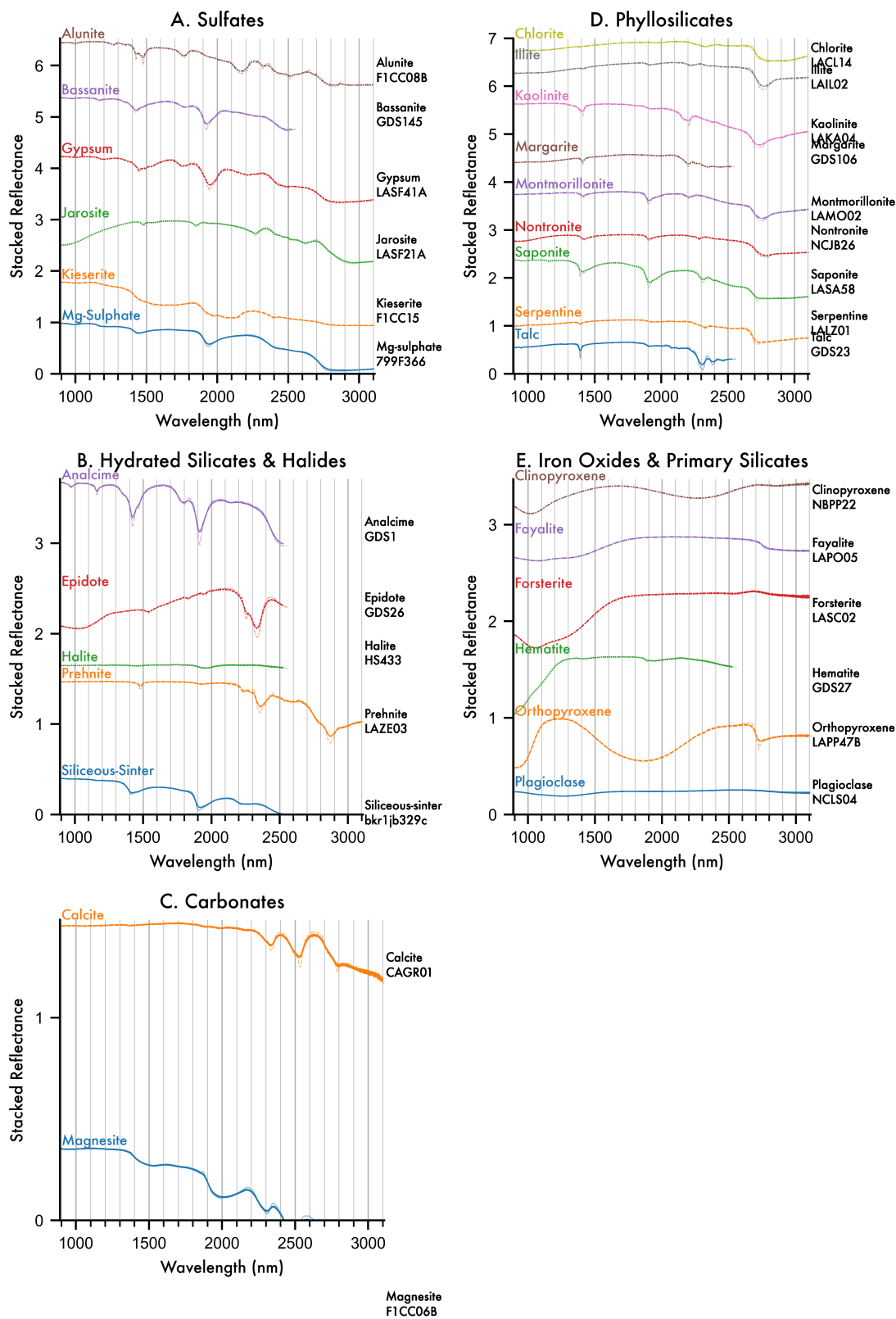
**Figure 7** Mean of Noisy resampled example spectrum for Illite, showing mean and standard deviation of 100 repeat samples according to the spectral signal-to-noise ratio illustrated in figure 3.B.

Here we find that the standard deviation is negligible.

Plotting out all of the groups, we again see that the noise, at this scale, is negligible.

```
[26]: axes = obs.plot_profiles(scope='categories', groupby='Species', stacked=True,
    ↪ hires_under=True, ci=False)
```

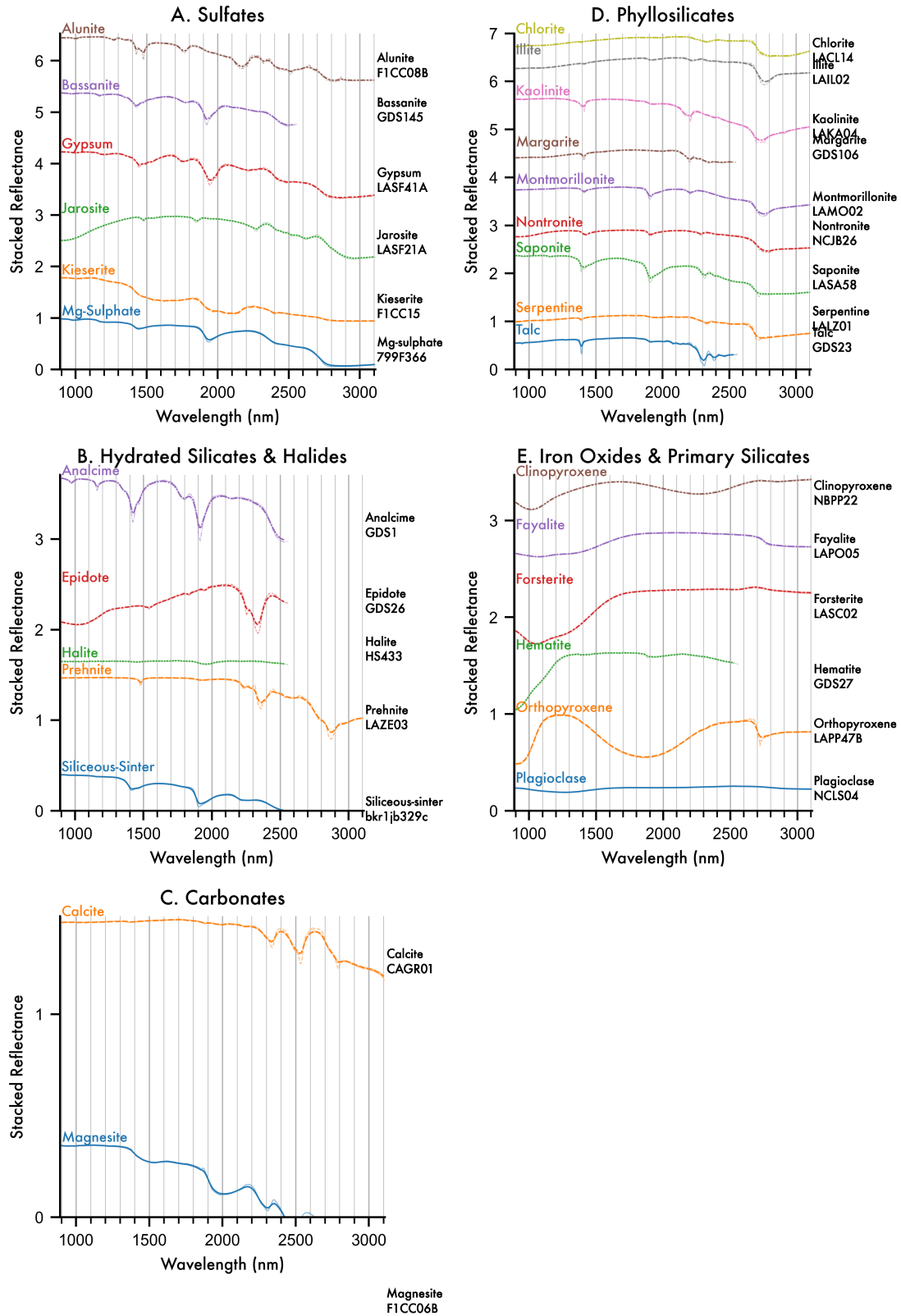
## Enfys Sampled Categories grouped by Species



Why are some entries end points below 0?

```
[27]: axes= obs.plot_profiles(scope='categories', groupby='Species', stacked=True,
    ↪ hires_under=True, ci=True)
```

## Enfys Sampled Categories grouped by Species

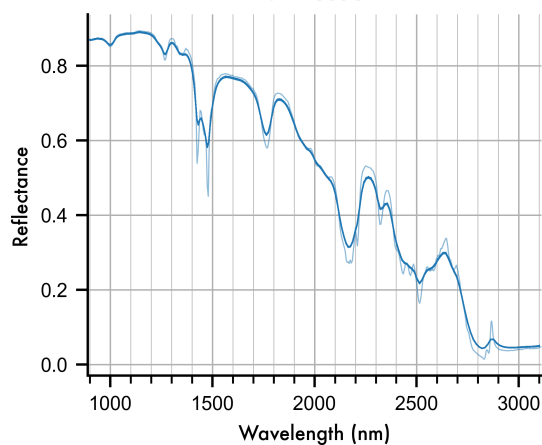


Let's plot out all of the profiles independently to clearly see the spectra.

```
[29]: axes = obs.plot_profiles(scope='samples', groupby='Species', stacked=False,
    ↪ hires_under=True, ci=False)
```

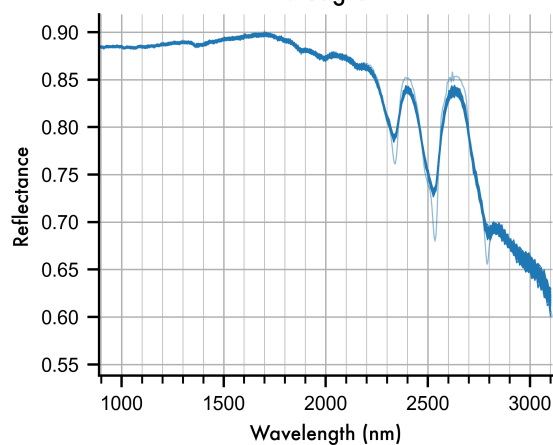
# Enfys Sampled Samples grouped by Species (1/5)

A. F1Cc08B



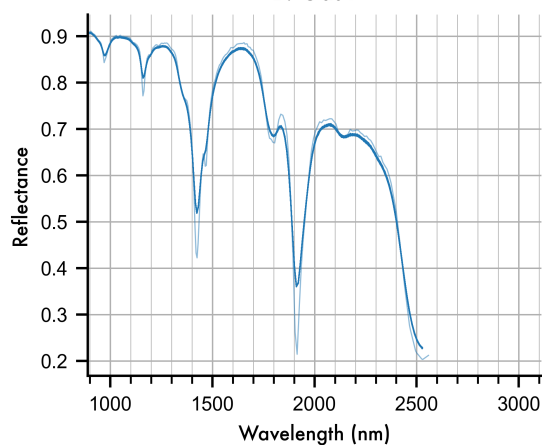
Alunite

D. Cagr01



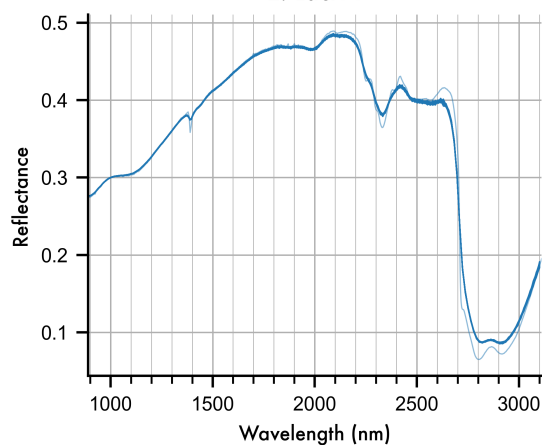
Calcite

B. Gds1



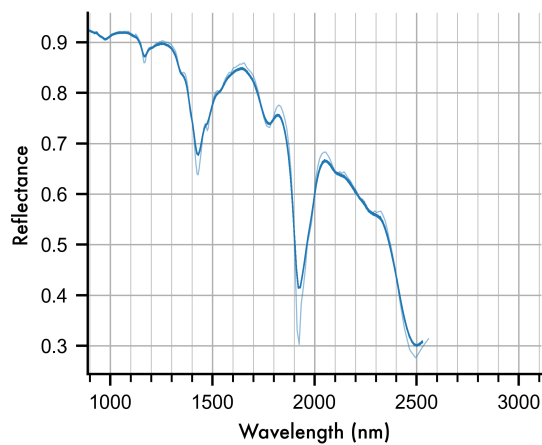
Analcime

E. Lac114



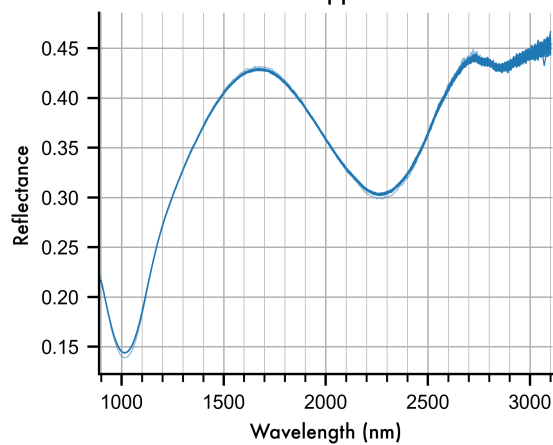
Chlorite

C. Gds145



Bassanite

F. Nbpp22

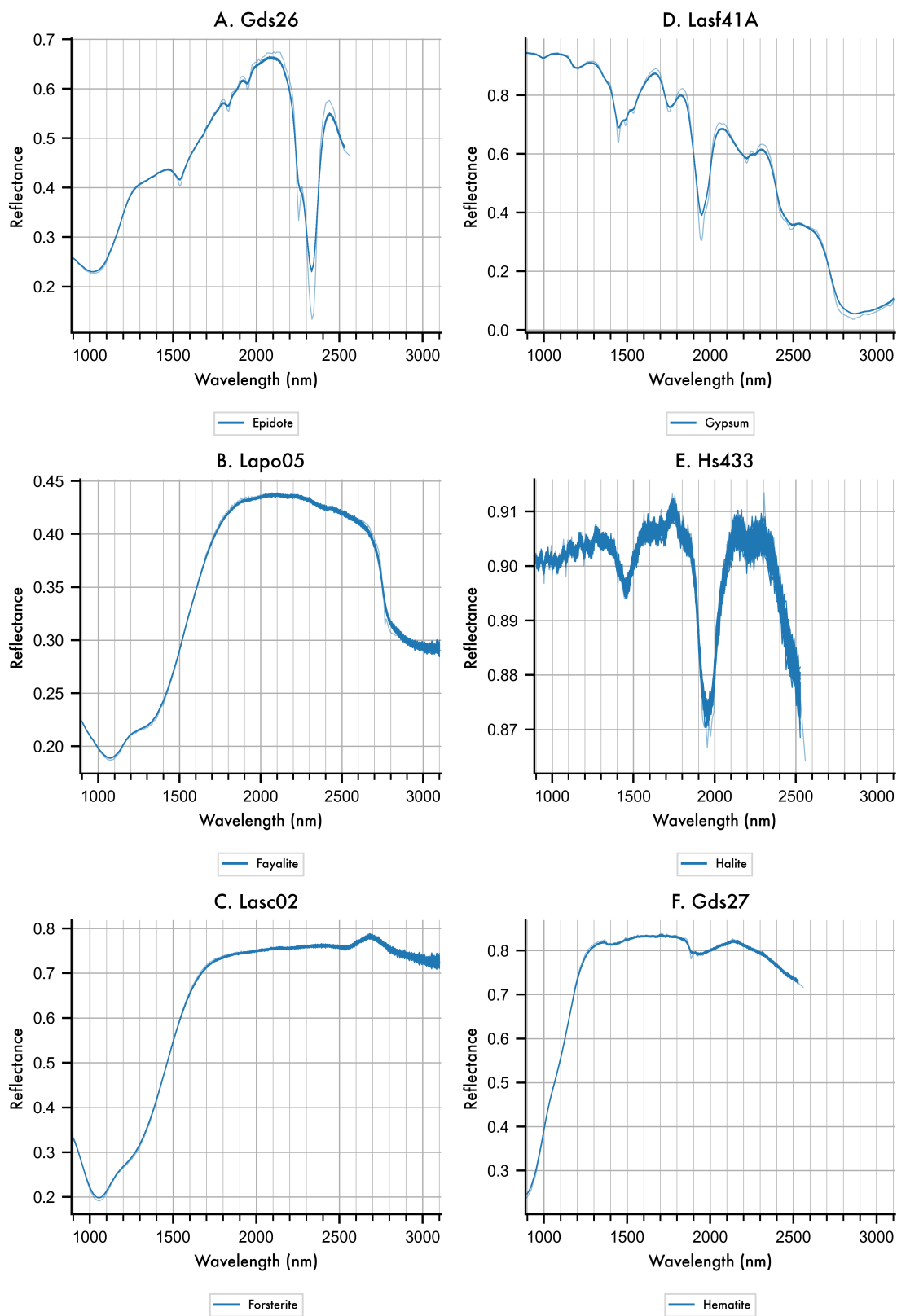


Clinopyroxene





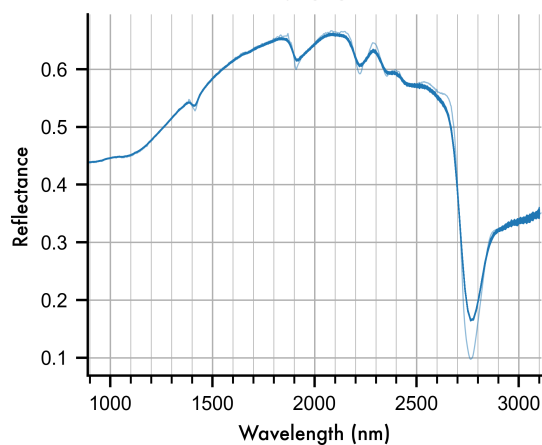
# Enfys Sampled Samples grouped by Species (2/5)



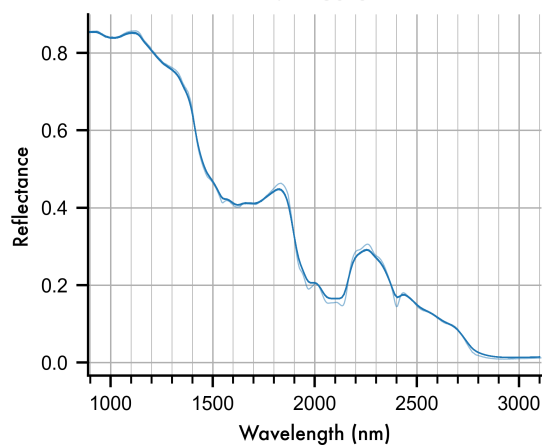


# Enfys Sampled Samples grouped by Species (3/5)

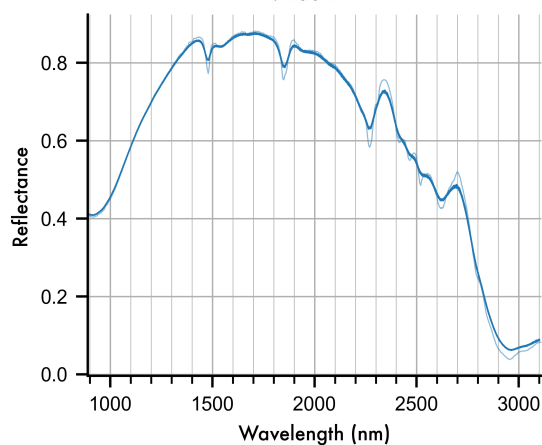
A. Lail02



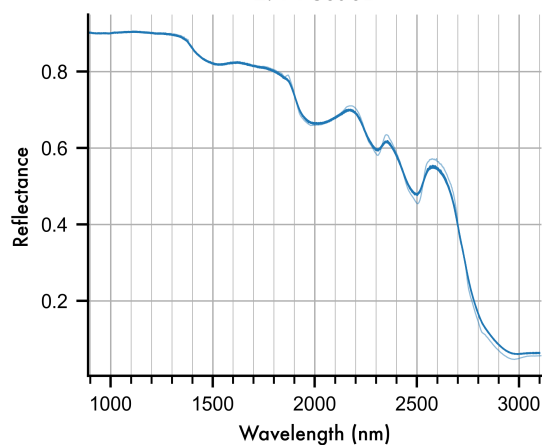
D. F1Cc15



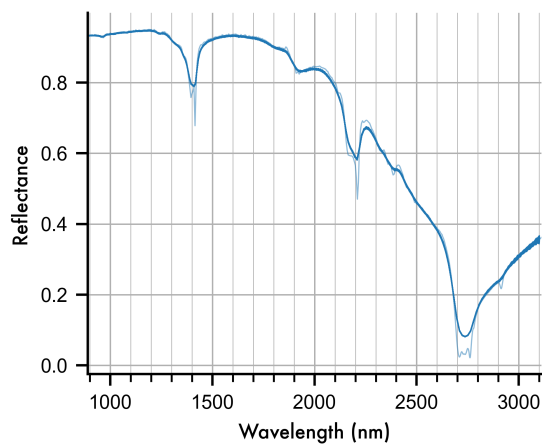
B. Lasf21A



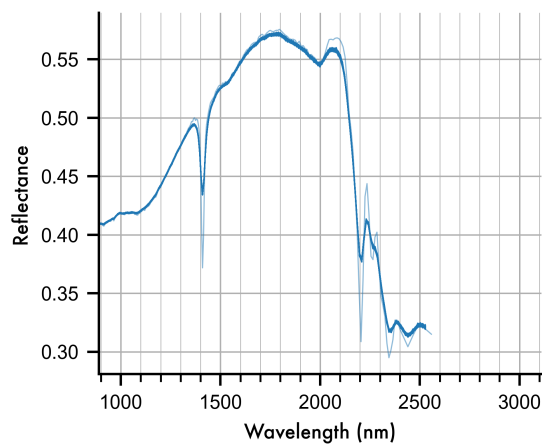
E. F1Cc06B



C. Laka04



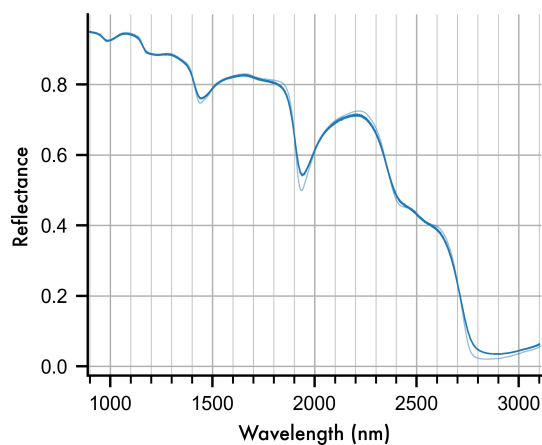
F. Gds106



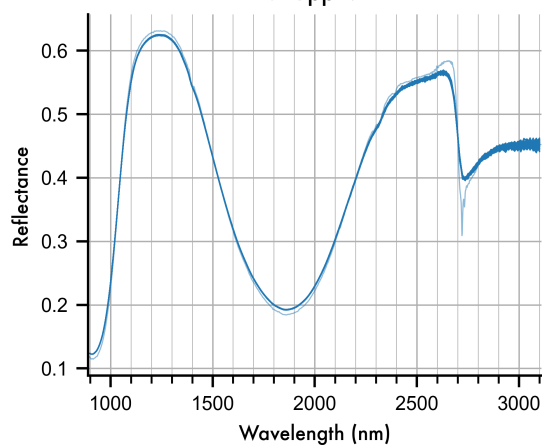


# Enfys Sampled Samples grouped by Species (4/5)

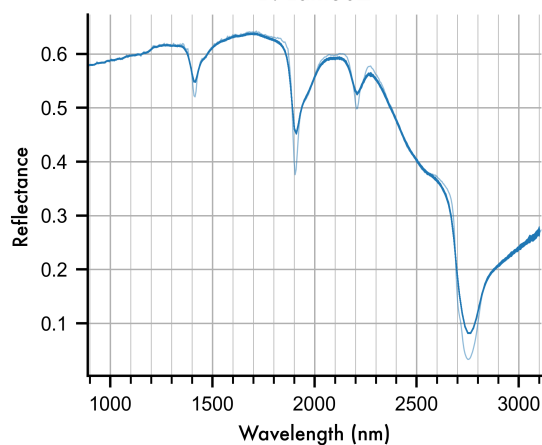
A. 799F366



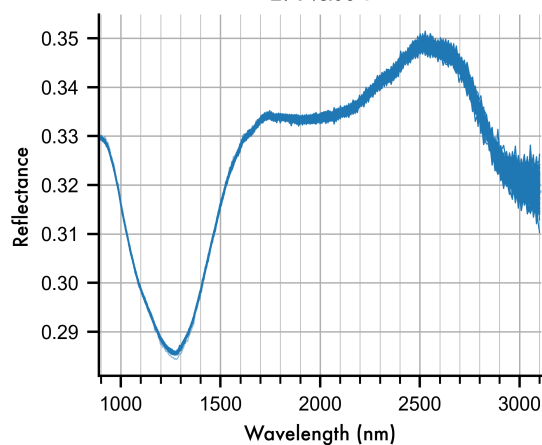
D. Lapp47B



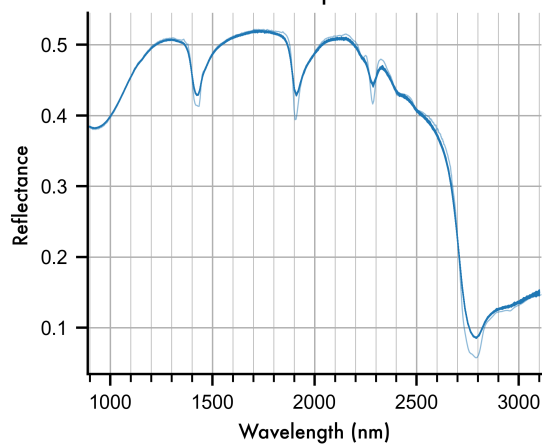
B. Lamo02



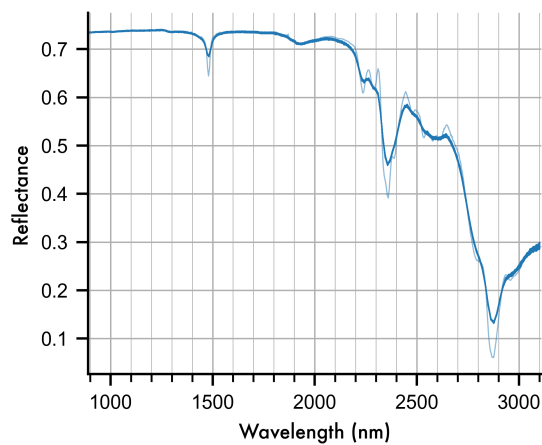
E. Ncls04



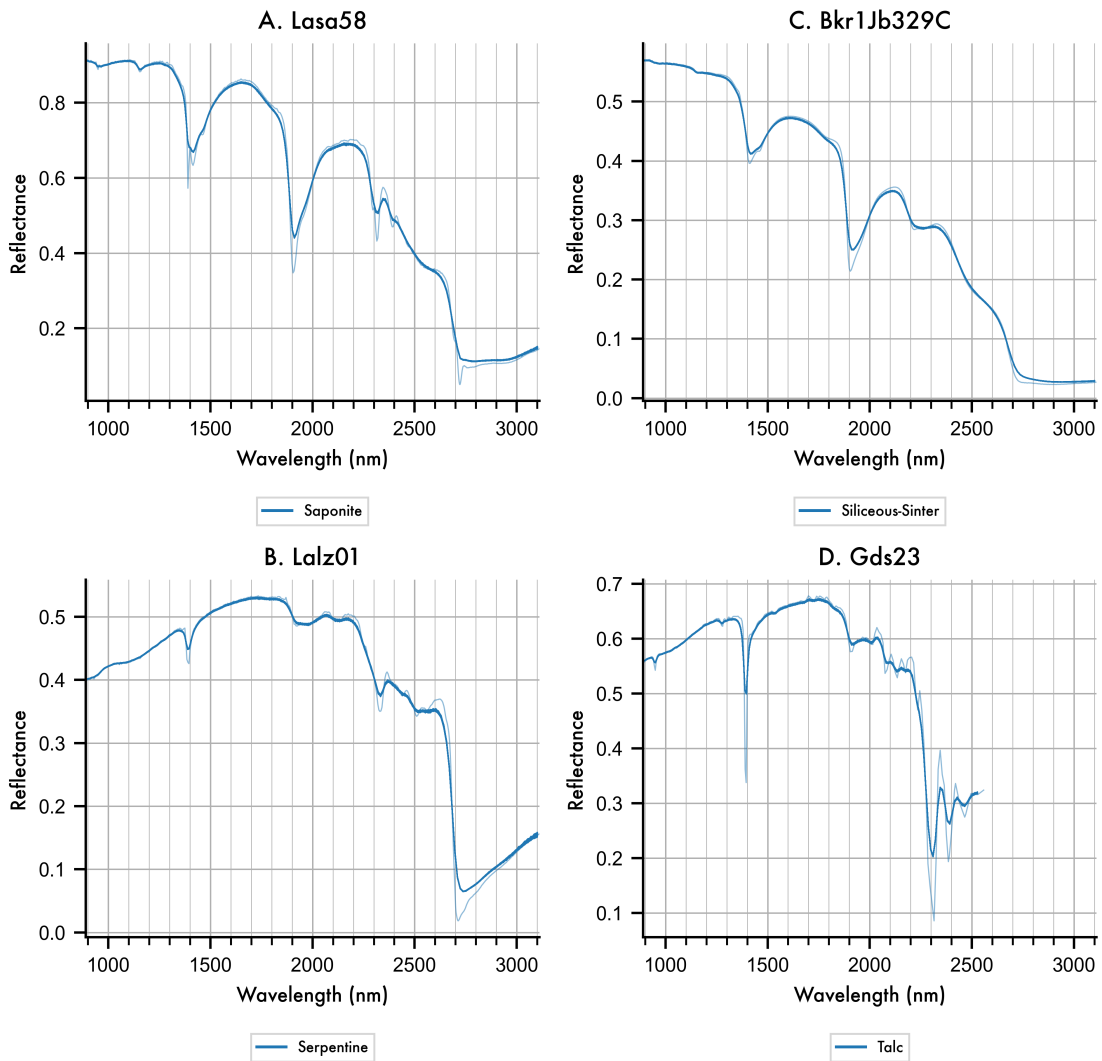
C. Ncjb26



F. Laze03



## Enfys Sampled Samples grouped by Species (5/5)



## 6 Discussion

Notes on what we found above, perhaps with some more analysis.

## 7 Conclusions

Concluding remarks and summary of findings.

## 8 References